

§3-2 The Product and Quotient Rules

Homework: 2,5,11,17,25,32,34,35,38(d),42,43

(I) Product Rule

$$(fg)' = f'g + fg'.$$

(II) Quotient Rule

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

Proof of (I) :

已知： f and g are differentiable.

求證：(I) holds.

$$\begin{aligned}(fg)'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{g(x+h)[f(x+h) - f(x)] + f(x)[g(x+h) - g(x)]}{h} \\ &\quad (\text{做了加一項減一項回來的動作})\end{aligned}$$

Since f and g are differentiable, and, hence continuous, the limit above exists and equals to

$$g(x)f'(x) + f(x)g'(x).$$

Remark (III) : $(fgh)' = f'gh + fg'h + fgh'$

Question : Why formulas such as (I) and (III) are “reasonable” ?

(考慮當 f, g, h 是多項式時的特例，為何(I)and (III) 是合理的)

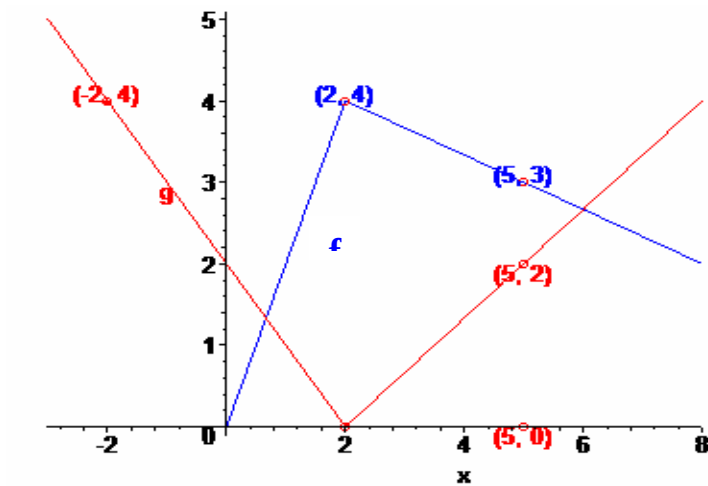
Example 1 : $f(x) = \frac{x - 3x\sqrt{x}}{\sqrt{x}}$, Find $f'(x)$.

Solution : $f(x) = x^{\frac{1}{2}} - 3x \Rightarrow f'(x) = \frac{1}{2}x^{-\frac{1}{2}} - 3$.

Example 2 : $h(2) = 4, h'(2) = -3$ Find $\frac{d}{dx}\left(\frac{h(x)}{x}\right)\bigg|_{x=2}$.

Solution : $\left. \frac{d}{dx} \left(\frac{h(x)}{x} \right) \right|_{x=2} = \left. \frac{h'(x) x - h(x)}{x^2} \right|_{x=2} = -\frac{5}{2}.$

Example 3 :



Let $u(x) = f(x)g(x)$, $v(x) = \frac{f(x)}{g(x)}.$

(a) Find $u'(1).$ (b) Find $v'(5).$

Solution : $u'(1) = f'(1)g(1) + f(1)g'(1)$

$$= 2 \cdot 1 + 2 \cdot (-1) = 0$$

$$v'(5) = \frac{f'(5)g(5) - f(5)g'(5)}{[g(5)]^2}$$

$$= \frac{\left(-\frac{1}{3}\right) \cdot 2 - 3 \cdot \frac{2}{3}}{2^2} = -\frac{2}{3}.$$